

tf.compat.v1.linalg.matrix_transpose

Stay organized with collections Save and categorize content based on your preferences.

https://www.tensorflow.org/api_docs/python/tf/compat/v1/linalg/matrix_transpose

Transposes last two dimensions of tensor a.

Function Signatures

```
tf.compat.v1.linalg.matrix_transpose( a,  
name='matrix_transpose', conjugate=False )
```

Code Examples

```
tf.compat.v1.linalg.matrix_transpose(  
a, name='matrix_transpose', conjugate=False  
)
```

```
tf.compat.v1.linalg.matrix_transpose(  
a, name='matrix_transpose', conjugate=False  
)
```

```

x = tf.constant([[1, 2, 3], [4, 5, 6]])
tf.linalg.matrix_transpose(x) # [[1, 4],
# [2, 5],
# [3, 6]]
x = tf.constant([[1 + 1j, 2 + 2j, 3 + 3j],
[4 + 4j, 5 + 5j, 6 + 6j]])
tf.linalg.matrix_transpose(x, conjugate=True) # [[1 - 1j, 4 - 4j],
# [2 - 2j, 5 - 5j],
# [3 - 3j, 6 - 6j]]
# Matrix with two batch dimensions.
# x.shape is [1, 2, 3, 4]
# tf.linalg.matrix_transpose(x) is shape [1, 2, 4, 3]

```

```

x = tf.constant([[1, 2, 3], [4, 5, 6]])
tf.linalg.matrix_transpose(x) # [[1, 4],
# [2, 5],
# [3, 6]]
x = tf.constant([[1 + 1j, 2 + 2j, 3 + 3j],
[4 + 4j, 5 + 5j, 6 + 6j]])
tf.linalg.matrix_transpose(x, conjugate=True) # [[1 - 1j, 4 - 4j],
# [2 - 2j, 5 - 5j],
# [3 - 3j, 6 - 6j]]
# Matrix with two batch dimensions.
# x.shape is [1, 2, 3, 4]
# tf.linalg.matrix_transpose(x) is shape [1, 2, 4, 3]

```

```

# Good! Transpose is taken at minimal additional cost.
tf.matmul(matrix, b, transpose_b=True)
# Inefficient!
tf.matmul(matrix, tf.linalg.matrix_transpose(b))

```

```

# Good! Transpose is taken at minimal additional cost.
tf.matmul(matrix, b, transpose_b=True)
# Inefficient!
tf.matmul(matrix, tf.linalg.matrix_transpose(b))

```

Documentation

Transposes last two dimensions of tensor a.

View aliases

Compat aliases for migration

See

Migration guide for
more details.

`tf.compat.v1.linalg.transpose`

For example:

Note that `tf.matmul` provides kwargs allowing for transpose of arguments. This is done with minimal cost, and is preferable to using this function. E.g.

Returns

Raises

numpy compatibility

In numpy transposes are memory-efficient constant time operations as they simply return a new view of the same data with adjusted strides.

TensorFlow does not support strides, `linalg.matrix_transpose` returns a new tensor with the items permuted.

